LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	34
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principle of operation of powered control (electro hydraulic servo motors)
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principle of operation of powered control (electro hydraulic servo motors).
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Servo Motor operation using PPT schematics; Safety: Do not touch live electrical components; Supervision: Instructor monitors trainee observation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) How does a manual control system differ from a powered control system?
(c) What is the role of the servo motor in the flight control system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of AFCS • Components of Electro Hydraulic Servo Motor • Signal flow from pilot input to actuator • Hydraulic power supply requirements • Feedback loop mechanism in servo motors	L & Ds	PPT, IB	15 min
(b)	• Operation during normal flight conditions • Response to pilot command inputs • Automatic trim and stabilization functions • Failure modes and system redundancy • Interaction with flight control computers	L & Ds	PPT, PDF	20 min
(c)	• Maintenance checks for servo motors • Troubleshooting common hydraulic leaks • Electrical fault diagnosis procedures • Testing servo response times • Record keeping for AFCS inspections	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the sequence of events when a pilot inputs a control command.
(b) Describe how the feedback loop ensures accurate servo motor positioning.
(c) Identify two potential failure modes in an electro hydraulic servo system.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of servo motors discussed in this lesson apply to the automatic trim control system covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Application of servo motors in automatic trim control.	Refer to the master syllabus reference list	LP 35

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min